

External Data Driven Estimation of Atmospheric Pollutants from Sub THz ISAC Channel Signatures

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Abstract—Integrated sensing and communication (ISAC) has been proposed as a way to reuse communication waveforms for environmental sensing in future non terrestrial networks. This paper studies an early external data driven pipeline for estimating atmospheric pollutants from sub THz channel amplitude signatures. Unlike a fully toy simulation, the proposed benchmark uses HITRAN spectroscopic line data in the 60 to 400 GHz band and real pollutant concentration records from the UCI Beijing Multi Site Air Quality dataset. Channel state information amplitudes are still generated by a simplified forward model, so the results should be interpreted as simulation based evidence rather than field validation. A template projection followed by Ridge regression achieves the best test performance with a mean normalized RMSE of 0.0779 and a mean R^2 of 0.945 across CO, O₃, SO₂, NO₂, PM2.5, and PM10. The results suggest that physically informed spectral compression is a promising baseline before moving to higher complexity machine learning models.

Index Terms—Integrated sensing and communication, sub THz, non terrestrial networks, HITRAN, air quality, atmospheric sensing, machine learning.

I. Introduction

Future sixth generation networks are expected to include non terrestrial links, high carrier frequencies, and integrated sensing and communication capabilities. In sub THz bands, molecular absorption and particle scattering are usually treated as impairments. The same effects may also encode useful environmental information. This motivates the question of whether a satellite downlink can infer atmospheric pollutant concentrations from wideband channel state information (CSI) amplitudes.

Prior work has explored sub THz satellite communication with differential absorption radar concepts [1], THz attenuation modeling across space air ground media [2], and opportunistic THz ISAC for environmental or space sensing [3], [4]. This paper does not claim field validation. Instead, it builds a reproducible intermediate benchmark using external spectroscopic and pollution data.

II. External Data Sources

A. HITRAN Line Data

The spectroscopic source is HITRAN accessed through the HAPI interface [5]. The frequency range is 60 to 400 GHz, equivalent to 2.001 to 13.343 cm⁻¹. The current study downloads the main isotopologue for CO, O₃, SO₂, NO₂, H₂O, and O₂. H₂O and O₂ are treated as nuisance background components.

B. Air Quality Records

The pollutant labels come from the UCI Beijing Multi Site Air Quality dataset [6]. The dataset contains hourly air quality and meteorological measurements from 12 monitoring sites between March 2013 and February 2017. This study uses PM2.5, PM10, SO₂, NO₂, CO, and O₃ after removing rows with missing required values.

III. Forward Model

CSI amplitudes are generated from real pollutant records using a frequency grid of 256 subcarriers between 60 and 400 GHz. For gas g , a normalized HITRAN template $T_g(f)$ is computed by summing Lorentzian line profiles weighted by relative HITRAN line intensity:

$$T_g(f) = \frac{1}{Z_g} \sum_{\ell \in g} \frac{s_\ell}{s_{g,\max}} \frac{1}{1 + ((f - f_\ell)/\gamma_\ell)^2}, \quad (1)$$

where f_ℓ is the HITRAN line frequency, s_ℓ is line intensity, γ_ℓ is the air broadened half width converted to GHz, and Z_g normalizes the template peak.

Particulate matter is represented using smooth Rayleigh type trends. The attenuation spectrum for record i is

$$S_i(f) = \sum_g c_{i,g} \alpha_g T_g(f) + c_{i,\text{PM2.5}} \alpha_{2.5} T_{2.5}(f) + c_{i,\text{PM10}} \alpha_{10} T_{10}(f) + B(f), \quad (2)$$

$$A_i(f) = p_i S_i(f) + n_i(f). \quad (3)$$

where p_i is the slant path factor from satellite elevation, $B(f)$ is an H₂O/O₂ nuisance background, and $n_i(f)$ is additive measurement noise in dB. The constants α_g scale each pollutant by the 95th percentile of its observed concentration distribution.

IV. Estimators

Several feature sets were evaluated:

- path normalized spectra,
- mean removed path normalized spectra,
- path normalized spectra with elevation and SNR context,
- HITRAN template projection coefficients,
- a hybrid of spectra and template coefficients.

The model set includes linear regression, Ridge regression, partial least squares, k nearest neighbors, random

TABLE I
Best External Data Driven Benchmark Results

Config.	mNRMSE	Mean R^2	Max NRMSE
Tpl + Ridge 17.8	0.0779	0.9452	0.0946
Tpl + Ridge 10	0.0779	0.9452	0.0946
Path + Ridge 10	0.0797	0.9428	0.0975

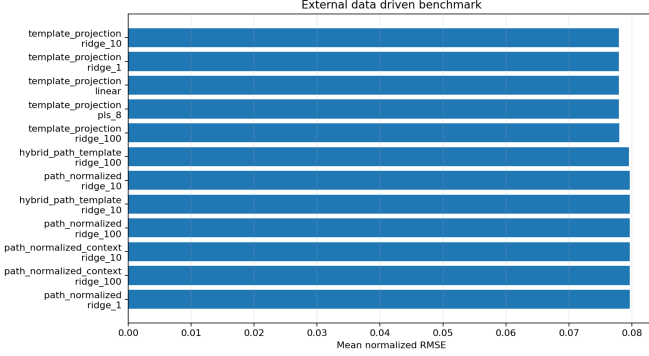


Fig. 1. Top external data driven estimator configurations ranked by mean normalized RMSE.

forest, extra trees, and a HITRAN template least squares estimator. All reported metrics use a held out test split.

V. Results

Table I reports the best configurations. Mean normalized RMSE averages per target RMSE after normalizing each target by its 5th to 95th percentile range.

The best configuration uses HITRAN template projection followed by Ridge regression. The improvement over the raw path normalized spectrum is modest but stable. A focused Ridge sweep shows a flat optimum around $\alpha = 17.8$, indicating that the result is not sensitive to exact regularization.

VI. Limitations

This study has several important limitations. First, CSI is simulated rather than measured. Second, HITRAN lines are currently used as normalized templates rather than full calibrated absorption coefficients. Third, PM attenuation is modeled by simplified smooth Rayleigh type trends. Fourth, only the main isotopologue is used for each molecule. Fifth, the forward model does not yet perform layer by layer atmospheric integration over pressure, temperature, and humidity.

VII. Conclusion

An external data driven benchmark was built for sub THz ISAC atmospheric monitoring. Using HITRAN line data and real air quality records, a template projection plus Ridge estimator achieved a mean normalized RMSE of 0.0779 and mean R^2 of 0.945 on held out samples. The next step is to replace normalized templates with HAPI absorption coefficients, implement a more rigorous PM scattering model, and derive estimation bounds.

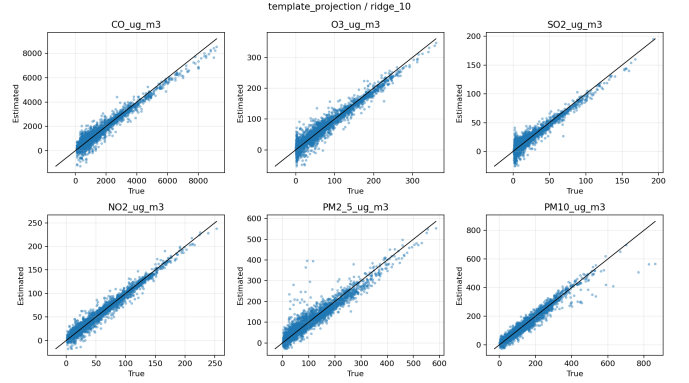


Fig. 2. Held out predictions for the best external data driven estimator.

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